FIELD GUIDE — NETWORK CAPTURE

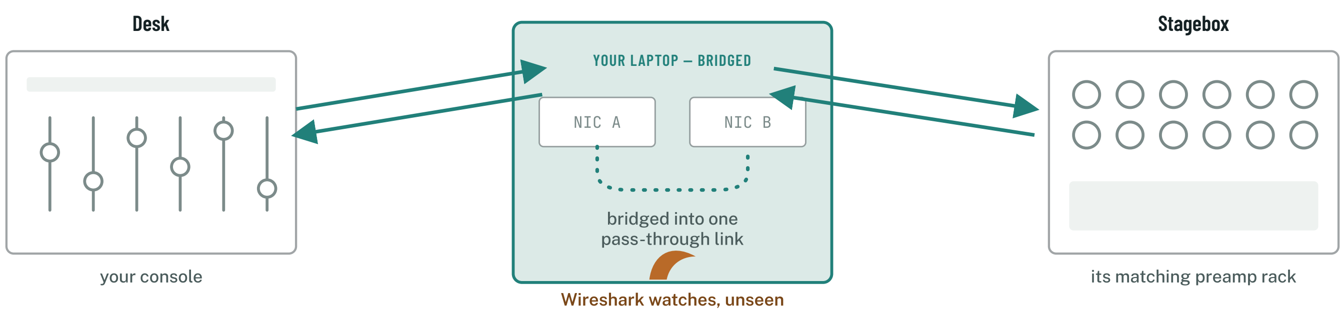
Watching Two Boxes Talk

A step-by-step guide to recording the private conversation between a mixing desk and its stagebox — using a laptop, two network cables, and one free piece of software. No specialist network gear required. This edition covers bridging on macOS.

NO TAP NEEDED | NO MIRROR SWITCH NEEDED | ~20 MINUTES | BENCH SETUP ONLY

THIS EDITION: Windows | **macOS** | Linux

DIAGRAM — THE IDEA IN ONE PICTURE 01



Every packet that would normally jump straight from desk to stagebox now physically passes through your laptop first — like fitting a clear length of pipe into a hose.

The whole trick: sit your laptop *in* the cable run, tell its two network ports to quietly pass everything through, and let Wireshark watch what goes by.

Every desk-and-stagebox pair from the same manufacturer has a private handshake — how they find each other, pair up, and agree on gain and phantom power — that's never been written down publicly. To build tools that can join that conversation from outside, we first have to record it happening between two devices that already trust each other.

> **Bench test, not showtime**
>
> Do this on a spare desk-and-stagebox pair, before or after a show — never on a live network mid-event. Inserting anything into a production Dante network, even briefly, isn't worth the risk.

§1 What you'll need

Nothing here is specialist. If you already run Dante gear, you likely own everything but the adapter cable — and that's a few pounds at any computer shop.

The desk Your console. REQUIRED	**Its matching stagebox** Same brand and family as the desk — you want the pairing they already trust. REQUIRED	**Two Ethernet cables** The ordinary ones you'd use anyway. REQUIRED
A laptop with two network ports A built-in Ethernet port (or Thunderbolt adapter) plus a cheap USB-to-Ethernet adapter covers almost any Mac. REQUIRED	**Wireshark, installed** Free at `wireshark.org`. The installer also installs ChmodBPF — say yes when it asks, or capture will need `sudo` every time. REQUIRED	**A mirroring switch** Not needed for the method above — but there's now a ~£25 switch that does it, and one situation where mirroring is the *only* way. See §3. OPTIONAL

§2 The setup, step by step

Nine steps, done in order, from cabling to a saved file ready to hand over.

1 Wire it up

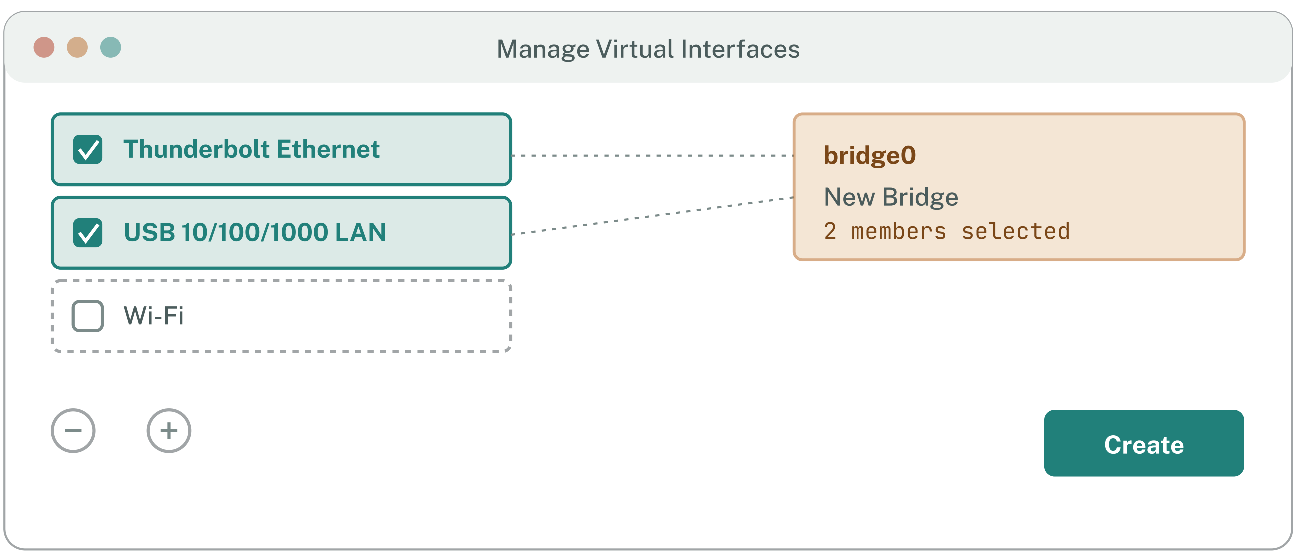
Run a cable from the desk's Ethernet port to **NIC A** on your laptop. Run a second cable from **NIC B** on your laptop to the stagebox's Ethernet port. That's it — no switch in between, your laptop *is* the connection now.

2 Bridge your two network ports

macOS can do this too — it's just tucked a little deeper than Windows, in a menu most people never open. Open `System Settings → Network`, click the `···` button at the bottom of the interface list (a gear icon on older macOS) and choose `Manage Virtual Interfaces…`.

Click `+ → New Bridge`, tick both of your Ethernet interfaces — the built-in port and your USB-to-Ethernet adapter — and click `Create`.

MOCKUP — MACOS MANAGE VIRTUAL INTERFACES



Tick both Ethernet interfaces, leave Wi-Fi out, and create the bridge. macOS names it `bridge0`.

Using Windows or Linux instead? There's a matching guide for each — the bridging step works differently, but everything else here is the same.

3 Open Wireshark and pick your interface

Look for `bridge0` in Wireshark's interface list — that's the one to capture on.

MOCKUP — WIRESHARK INTERFACE LIST

The Wireshark Network Analyzer

bridge0	← capture here
en0 — Thunderbolt Ethernet	idle
en5 — USB 10/100/1000 LAN	idle
en1 — Wi-Fi	idle

Nothing on it yet — that's expected before anything's powered on.

If nothing shows up on `bridge0` once traffic is flowing, select both physical interfaces together instead and start both — Wireshark merges them into one capture automatically.

4 Start the capture first

Click the blue shark-fin `Start` button *before* powering anything on. This is what catches the very first discovery packets — the part we're actually missing today.

5 Power everything on

Turn the stagebox and desk on in whatever order you'd normally use, then wait about 30 seconds for them to find each other over the network.

6 Pair them, like normal

From the desk's own screen, connect to the stagebox exactly as you always would. No special settings — routine operation is exactly what we want recorded.

7

Wiggle a few things

Once, each: nudge a gain knob physically on the stagebox; nudge a different one from the desk's on-screen control; toggle phantom power on a channel; then disconnect the desk from the stagebox in software and reconnect it. That re-pairing moment is one of the most useful bits in the whole capture.

8

Stop and save

Hit the red `Stop` square, then `File → Save As`. Give it a name that says what it is — something like `ql1_tio1608_2026-07-14.pcapng` — and save.

9

Send it over, then tidy up

The `.pcapng` file on its own is enough — no need to trim or export anything yourself. Afterward, undo the bridge: `System Settings → Network → Manage Virtual Interfaces…`, select `bridge0`, and click the `-` button to delete it.

§3 Mirroring a port instead

You don't need any of this — the laptop-bridge method above works fine on its own, and it is still the fastest way to get us a capture. But there are two situations where mirroring a switch port is the nicer answer, and one where it is the only one.

> **The gear won't tolerate a laptop in the path.** Re-cabling means the stagebox vanishes for a few seconds, and not every desk takes that gracefully.

You'd rather not touch the audio path at all. A mirror is passive: desk and stagebox stay plugged in exactly where they were, and nothing you do can interrupt them.

One box is talking privately to another. A wireless receiver mounted on a console unicasts its telemetry straight to that console — an ordinary switch port sees none of it, and there may be no cable to insert yourself into. Mirroring is the only way to see that traffic at all.

The cheap route: a UniFi USW-Flex-Mini

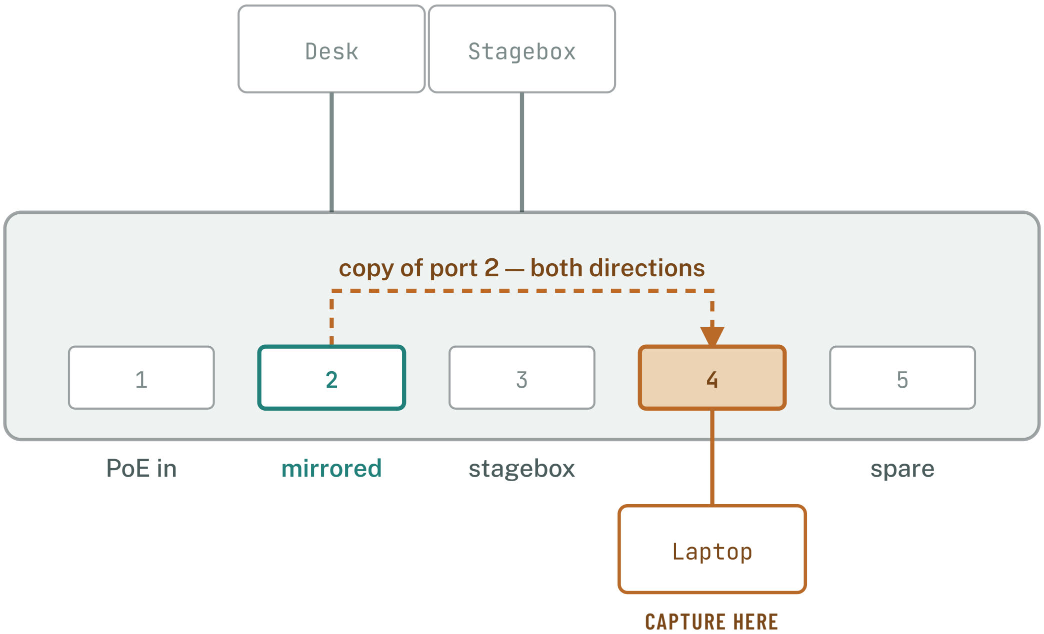
Owning a mirroring switch used to mean borrowing something from IT. It doesn't any more. The **UniFi USW-Flex-Mini** is a palm-sized five-port gigabit switch for about £25 / $30, and port mirroring is a listed feature — *“operation mode (switching or mirroring) per port”*, straight off Ubiquiti's own datasheet.

⚠ **The catch — read this before you buy one**

The Flex Mini has **no web interface of its own.** It has to be adopted by a UniFi Network controller before you can configure anything: a Dream Machine, a Cloud Key, or the free UniFi Network Application running on a Mac, PC, Raspberry Pi or in Docker. If you already run UniFi anywhere, you are two minutes from a mirror port. If you don't, standing a controller up is a bigger job than bridging your laptop — use the method above instead.

DIAGRAM — FIVE PORTS, ONE OF THEM A COPY

USW-FLEX-MINI



Power it over USB-C and all five ports stay free for gear; port 1 is the only one that takes PoE in, so over PoE it is already spoken for.

1

Wire it up

Desk into **port 2**, stagebox into **port 3**, your laptop into **port 4** — that last one is the capture port.

Leave your laptop on **Wi-Fi** for talking to the controller, and use its Ethernet port purely for capture. The note after step 4 explains why that isn't optional.

2

Turn port 4 into a mirror

In UniFi Network: `Devices` → click the Flex Mini → `Ports` → click **port 4**, the one your laptop is on → `Edit`.

Set `Operation` to **Mirroring**, set `Mirroring Port` to **2** — the desk's port — then `Apply Changes` and give the switch a few seconds to provision.

> You are always editing the port the copy comes **out of**, and then naming the port being **copied**. Newer UniFi Network versions wrap this in a port profile and move things around, but it is the same two questions in the same order.

3

Mirror the desk — and one port really is enough

UniFi's mirroring is strictly **one source port to one destination**. There is no “mirror everything onto port 4” here, and no CLI to sneak around it — the Flex Mini doesn't even offer SSH.

That turns out not to matter for this job. Mirroring a port copies traffic in **both directions**, so mirroring port 2 gives you everything the desk sends *and* everything the stagebox sends back to it. That is the entire conversation, from one port. You would only want more if a third device were joining in.

4

Capture

Start Wireshark on your laptop's ordinary Ethernet interface. There is no bridge to make here — no `bridge0` to create, and nothing to delete afterwards.

Then follow **steps 4 to 9** of §2 exactly as written: start the capture first, power everything on, pair them, wiggle a few things, stop, save, and send the file over. Only the tidy-up half of step 9 changes — set the port back to `Switching` rather than deleting a bridge. A port left in mirroring mode looks simply broken to whoever plugs into it next.

Two things to expect while it is running

Your laptop's Ethernet port goes deaf

A mirror destination only spits copies out — it won't carry a working network connection. Your laptop can't reach the controller, or the internet, through that port while mirroring is on. That is correct behaviour, not a fault, and it is why the controller needs to be reachable over Wi-Fi instead.

EXPECTED — NOT A FAULT

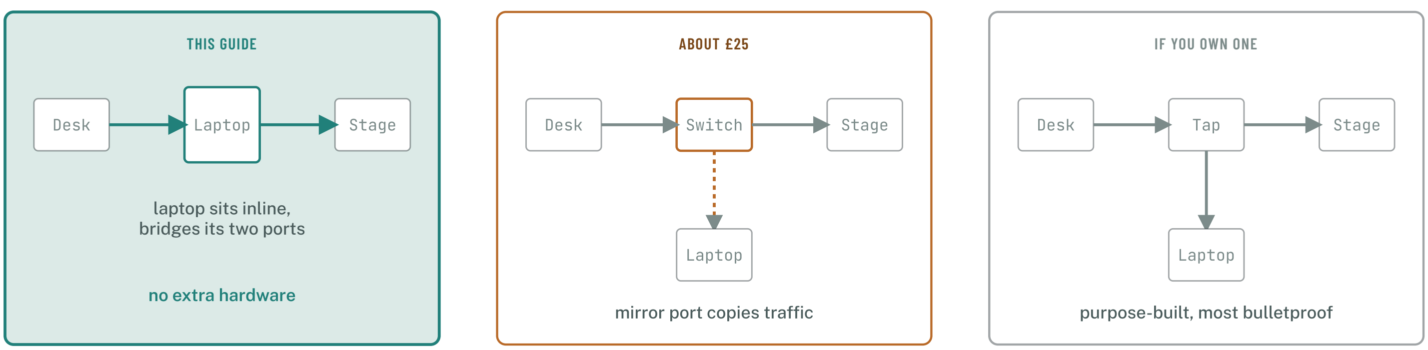
The file gets big, fast

If Dante *audio* is flowing through port 2, you are copying every one of those streams too — tens of megabytes per second. Before you start, in Wireshark: `Capture → Options → Output`, tick **Create a new file automatically** every **100 MB**, and tick **Use a ring buffer with 20 files**. You finish with the most recent 2 GB instead of a full disk.

SET THE RING BUFFER BEFORE YOU START

The other two options

DIAGRAM — THREE WAYS TO SEE THE SAME TRAFFIC



A managed switch you already own

Any switch with a mirror / SPAN port does the same job as the Flex Mini above, and most of them will mirror several source ports at once rather than just one. The menus differ per vendor; the two questions don't.

NEEDS ADMIN ACCESS TO THE SWITCH

Hardware network tap

A small dedicated box that sits inline between desk and stagebox, just like the laptop-bridge method, and quietly copies every packet out a third port to your laptop. The most bulletproof option, since it never routes the devices' whole conversation through a general-purpose computer.

EXTRA KIT TO BUY — NOT REQUIRED

§4 If something's not working

? I can't find "Manage Virtual Interfaces..."

It's under the `···` (or gear) button at the very bottom of the Network settings interface list — on older macOS versions it's a dropdown directly under the gear icon rather than a separate button.

? Wireshark asks for my password, or the interface list is empty

The ChmodBPF helper wasn't installed. Re-run the Wireshark `.pkg` installer and make sure the ChmodBPF step is ticked, then log out and back in.

? bridge0 shows no traffic

Capture on both individual physical interfaces at the same time instead — Wireshark merges them into a single, correctly-ordered capture automatically.

? The desk and stagebox won't find each other through the laptop

Check `System Settings → Network → Firewall` isn't blocking the multicast traffic discovery relies on, and try turning Wi-Fi off entirely so it can't interfere.

? There's no "Mirroring" option on the switch's ports

The switch is either not adopted yet, or running old firmware — check for a firmware update in UniFi Network and let it apply, then look again. If it still isn't there, fall back to the laptop-bridge method in §2.

? The mirror port is set up but Wireshark sees nothing

Check you edited the *laptop's* port and pointed it at the *desk's* port, and not the other way round — it is an easy one to get backwards, and backwards produces exactly this. Also check the laptop's Ethernet interface is up at all: some NICs report no link worth capturing on when nothing is being sent back to them.
